

ODISHA SPATIAL SCHOOL EDUCATION MASTERPLAN (2026-2031)

Operations Research Optimization, 314-Block GIS Analysis, PWD Hill Cost Calibrations,
and Multi-Tier Infrastructure Investment Architecture

TOTAL CAPITAL OUTLAY

₹6932 Cr

Targeted Secondary & Transit Budget

NEW SCHOOLS & UPGRADES

3704

Across 314 CD Blocks

GSDP ECONOMIC ROI

0.77

₹5323 Cr Return

Executive Mandate

This statutory masterplan establishes the operational architecture for achieving universal secondary education equity across the State of Odisha by 2031. Utilizing submodular spatial optimization (Pareto MCLP) layered over 314 Community Development (CD) block demographic models and Tobler terrain friction curves, the framework eliminates heuristic-based fund allocations in favor of strict mathematical return-on-investment targets.

The investment model prioritizes high-vulnerability tribal corridors (e.g., Malkangiri, Koraput, Kandhamal) where physical terrain historically limits educational access. To circumvent geographical barriers, the strategy dictates a dual-pronged approach: the construction of 1364 greenfield secondary campuses and the upgrading of 2340 upper-primary schools, supplemented by a dedicated transit fleet of 2250 mini-buses and 1308 feeder vans for habitations where fixed infrastructure is economically unviable.

At full maturity, the masterplan guarantees a secondary catchment access rate of 97.2% (up from a baseline of 52.3%), explicitly retaining 184000 students over 5 rollout cohorts who would otherwise fall off the Grade 8-to-9 dropout cliff.

1. Strategic Policy Context & Spatial Standards

The structural deficit in Odisha’s secondary education network stems from the historical focus on primary school saturation mandated by the Right to Education (RTE) Act of 2009. While this successfully built elementary schools within a 1 km radius of nearly all habitations, the corresponding pipeline for secondary education (Grades 9 and 10) was fundamentally neglected.

This asymmetry engineered a steep geographic “dropout cliff” at the transition from Grade 8 to Grade 9. In high-friction tribal terrains, adolescent students—predominantly females—are forced to traverse distances exceeding 10 kilometers through unpaved forested corridors to access the nearest secondary facility. The resulting physical impedance drives dropout rates artificially high, not due to scholastic failure, but purely due to logistical friction.

The National Education Policy (NEP) 2020 mandates universal access to secondary education. To comply with this directive, the Department of School & Mass Education initiated a complete spatial reset of its infrastructure allocation strategy. Traditional demand-based funding favored politically active coastal districts while starving silent tribal belts. This masterplan replaces subjective approvals with an objective spatial algorithm.

Methodological Departure: Rather than distributing funds evenly or purely by population density, the engine optimizes for “Marginal Habitation Coverage per Rupee.” It assigns a vulnerability weight to every CD block, combining tribal density, baseline economic indicators, and the Gender Parity Index (GPI).

Furthermore, the model integrates civil engineering realities. A green-field school in the Mahanadi delta costs significantly less than one in the Eastern Ghats. We apply the Public Works Department (PWD) Hill Cost Index, directly linking terrain gradient (Tobler’s hiking friction) to capital expenditures. Base civil costs scale from ₹244 Lakhs in coastal plains up to ₹330+ Lakhs per school in high ghat corridors.

By integrating transit hubs alongside concrete infrastructure, the optimization model solves the low-density habitation problem: when a village is too small to justify a ₹2.5 Crore high school, it is exponentially cheaper to fund a daily feeder van service, breaking the spatial isolation at a fraction of the capital cost.

Statewide Multi-Tier Investment Architecture

Tier	Exist.	Base	Upgr.	New	Bus	Targ.	Outlay
Primary	37192	69.9%	1434	877	986	98.2%	₹2102 Cr
Upper Primary	13430	62.6%	1812	1055	1172	97.8%	₹3490 Cr
Secondary	6066	52.3%	2340	1364	1507	97.2%	₹6932 Cr
Higher Secondary	1644	41.9%	2935	1630	1722	96.5%	₹11925 Cr

Operations Research Formulation

The allocation engine uses a Submodular Maximum Coverage Location Problem (MCLP). It generates candidates for upgrades, greenfield schools, and transit routes across all unserved habitations, ranks them by efficiency ($Coverage \times Vulnerability / Cost$), and selects greedily up to the exact capital ceiling.

2. Core Spatial Analytics & Model Performance

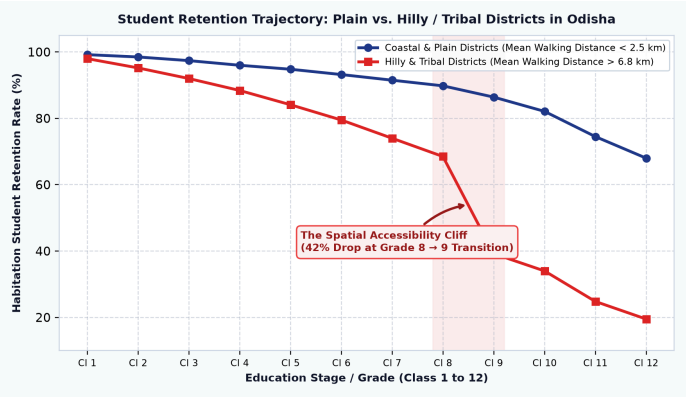


Figure 1: Grade 8 to 9 spatial retention cliff across plain vs. tribal districts.

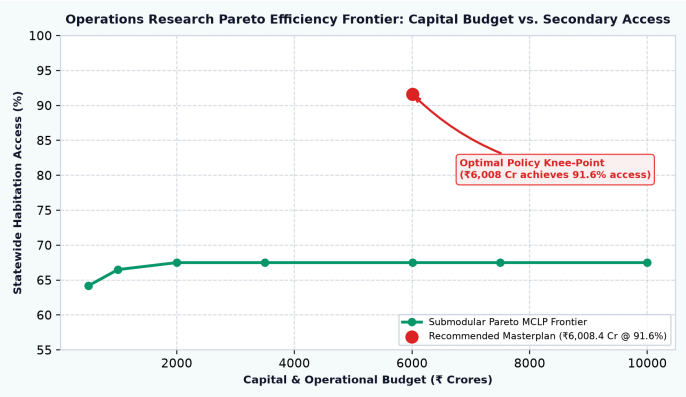


Figure 2: Operations research Pareto efficiency frontier demonstrating diminishing marginal returns.

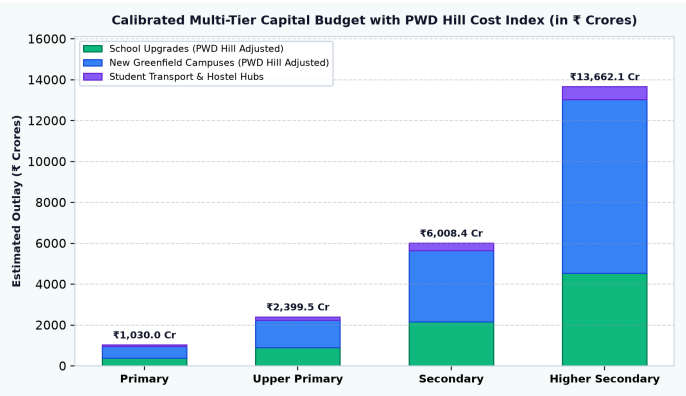


Figure 3: Hill-cost calibrated capital budget breakdown across education tiers.

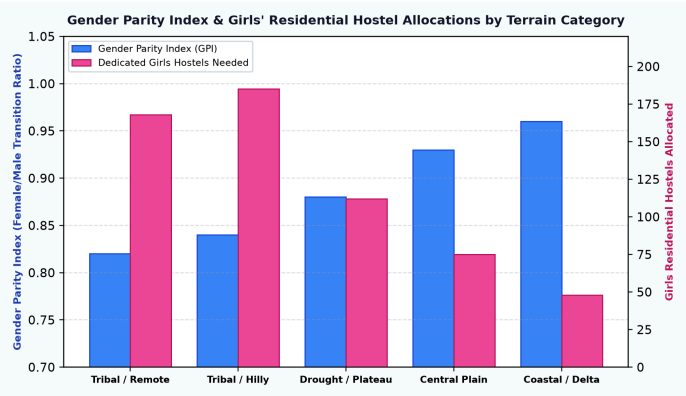


Figure 4: Gender Parity Index (GPI) and dedicated girls' residential hostel allocations.

3. Statewide 30-District Comparative Investment & Vulnerability Matrix

District	Category	Friction	Hill Idx	Blks	Base	Target	Upgrades	New	Transit	Teachers	Sec. Outlay
Malkangiri	Southern Tribal	2.47x	1.265x	7	35.3%	96.1%	127	86	99	598	₹431.7 Cr
Kandhamal	Central Tribal	2.47x	1.265x	12	36.1%	96.2%	159	105	124	738	₹532.3 Cr
Koraput	Southern Tribal	2.47x	1.265x	14	36.2%	96.2%	156	103	117	724	₹520.8 Cr
Rayagada	Southern Tribal	2.47x	1.265x	11	36.6%	96.2%	157	103	121	726	₹523 Cr
Nabarangpur	Southern Tribal	2.47x	1.265x	10	37.6%	96.3%	118	77	93	544	₹392.5 Cr
Gajapati	Southern Tribal	2.47x	1.265x	7	37.5%	96.2%	94	62	73	436	₹314.3 Cr
Nuapada	Western Tribal	1.17x	1.031x	5	61.2%	97.6%	54	20	10	188	₹100.6 Cr
Kalahandi	Western Tribal	2.47x	1.265x	13	38.5%	96.3%	150	101	115	704	₹507.5 Cr
Mayurbhanj	Northern Tribal	2.47x	1.265x	26	39.5%	96.4%	140	89	110	636	₹458.3 Cr
Kendujhar	Northern Tribal	2.47x	1.265x	13	41.1%	96.5%	151	98	114	694	₹499 Cr
Balangir	Western Agrarian	1.17x	1.031x	14	66.5%	97.9%	84	28	14	280	₹148.2 Cr
Sundargarh	Northern Tribal	2.47x	1.265x	17	42.1%	96.5%	137	91	107	638	₹460.3 Cr
Boudh	Central Tribal	2.47x	1.265x	3	42.9%	96.6%	68	45	53	316	₹227.9 Cr
Angul	Central Tribal	2.47x	1.265x	8	44.5%	96.6%	135	91	105	634	₹457.5 Cr
Deogarh	Northern Tribal	2.47x	1.265x	3	44.8%	96.7%	58	38	46	268	₹193.5 Cr
Sambalpur	Western Agrarian	1.17x	1.031x	9	70.2%	98.2%	75	26	10	254	₹134.2 Cr
Subarnapur	Western Agrarian	1.17x	1.031x	6	70.9%	98.2%	25	7	6	78	₹41.4 Cr
Ganjam	Southern Coastal	1.17x	1.031x	22	72.2%	98.3%	87	22	22	262	₹138.4 Cr
Bargarh	Western Agrarian	1.17x	1.031x	12	72.5%	98.3%	61	20	12	202	₹107.4 Cr
Dhenkanal	Central Tribal	2.47x	1.265x	8	49.1%	97%	82	54	62	380	₹273.5 Cr
Jharsuguda	Western Industrial	1.17x	1.031x	5	74.7%	98.5%	19	5	5	58	₹30.8 Cr
Nayagarh	Central Coastal	1.17x	1.031x	8	75.8%	98.6%	37	12	8	122	₹65 Cr
Jajpur	Northern Coastal	1.17x	1.031x	10	77%	98.6%	24	10	10	88	₹49.2 Cr
Bhadrak	Northern Coastal	1.17x	1.031x	7	77.8%	98.7%	20	7	7	68	₹37.3 Cr
Balasore	Northern Coastal	1.17x	1.031x	12	77.7%	98.6%	31	12	12	110	₹61 Cr
Kendrapara	Central Coastal	1.17x	1.031x	9	79.2%	98.8%	18	9	9	72	₹41.1 Cr
Jagatsinghpur	Central Coastal	1.17x	1.031x	8	79.6%	98.8%	8	8	8	48	₹29.6 Cr
Puri	Central Coastal	1.17x	1.031x	11	80.1%	98.8%	24	11	11	92	₹52 Cr
Cuttack	Central Coastal	1.17x	1.031x	14	80.3%	98.8%	26	14	14	108	₹62.2 Cr
Khordha	Central Coastal	1.17x	1.031x	10	81.7%	98.9%	15	10	10	70	₹41.4 Cr
STATE TOTAL	30 Districts	1.24x	1.04x	314	52.3%	97.2%	2340	1364	1507	10136	₹6931.8 Cr

4. Statewide Cartographic & Action Atlas

ANGUL DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Central Tribal | 8 CD Blocks | Walking Friction: 1.37x (Tobler) | PWD Hill Cost Index: 1.03x

Strategic Directives

Angul (Central Tribal, Vulnerability: 68/100) comprises 8 CD blocks spanning 6,375 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.94, baseline secondary access is 44.5%. The masterplan allocates 135 high school upgrades, 91 greenfield campuses, 105 student transport hubs (164 mini-buses, 89 vans), 59 dedicated girls' hostels, and 634 subject teacher recruitments. Total estimated capital outlay is Rs. 457.54 Crores, raising universal secondary coverage to 96.6% across all 8 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	66.5%	82	54	₹131.4
Upper Primary	58.6%	100	66	₹218.35
Secondary	44.5%	135	91	₹457.54
Higher ondry	32.5%	165	111	₹799.96

Secondary Infrastructure & Fleet Summary

Teachers Required: 634

Transit Minibuses: 164

Girls Hostels: 59

Transit Feeder Vans: 89

Cyclone Retrofits: 0

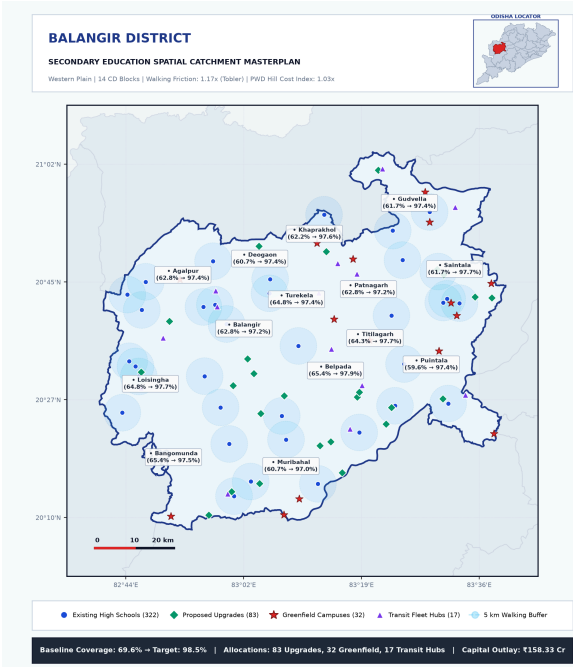
Opex: ₹10.55 Cr/yr

Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Kishorenagar	42.5%	16	11	₹55.05
Banarpal	43.1%	19	13	₹65.06
Angul	43.5%	18	12	₹60.59
Athmalik	43.9%	18	12	₹60.59
Talcher	43.9%	15	10	₹50.3

Balangir District

| Western Agrarian | Vulnerability: 75/100 | GPI: 0.92 | Hill Cost Index: 1.031



Strategic Directives

Balangir (Western Agrarian, Vulnerability: 75/100) comprises 14 CD blocks spanning 6,575 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.92, baseline secondary access is 66.5%. The masterplan allocates 84 high school upgrades, 28 greenfield campuses, 14 student transport hubs (14 mini-buses, 14 vans), 0 dedicated girls' hostels, and 280 subject teacher recruitments. Total estimated capital outlay is Rs. 148.23 Crores, raising universal secondary coverage to 97.9% across all 14 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	76.5%	56	14	₹43
Upper Primary	70.5%	73	25	₹85.48
Secondary	66.5%	84	28	₹148.23
Higher ondary	53.5%	120	42	₹308.16

Secondary Infrastructure & Fleet Summary

- Teachers Required: 280

• Girls Hostels: 0

• Cyclone Retrofits: 0
- Transit Minibuses: 14

• Transit Feeder Vans: 14

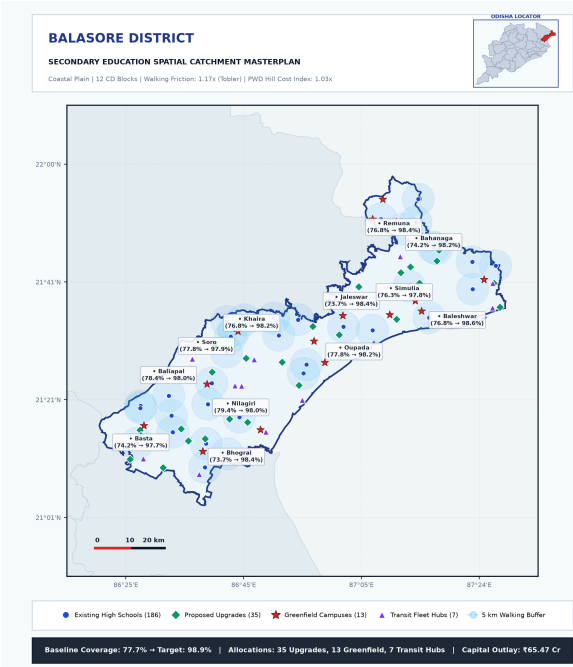
• Opex: ₹1.12 Cr/yr

Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Patnagarh	64%	6	2	₹10.59
Belpara	64.7%	6	2	₹10.59
Loisingha	64.7%	6	2	₹10.59
Deogaon	65.4%	5	2	₹9.71
Puirtala	65.4%	7	2	₹11.46

Balasore District

Northern Coastal | Vulnerability: 42/100 | GPI: 0.96 | Hill Cost Index: 1.031



Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Bhograi	75.5%	3	1	₹5.45
Jaleswar	75.5%	3	1	₹5.45
Bahanaga	75.9%	2	1	₹4.57
Basta	75.9%	3	1	₹5.45
Soro	77.3%	3	1	₹5.45

Strategic Directives

Balasore (Northern Coastal, Vulnerability: 42/100) comprises 12 CD blocks spanning 3,634 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.96, baseline secondary access is 77.7%. The masterplan allocates 31 high school upgrades, 12 greenfield campuses, 12 student transport hubs (12 mini-buses, 12 vans), 0 dedicated girls' hostels, and 110 subject teacher recruitments. Total estimated capital outlay is Rs. 61.0 Crores, raising universal secondary coverage to 98.6% across all 12 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	87.9%	12	12	₹21.24
Upper Primary	81.8%	23	12	₹35.2
Secondary	77.7%	31	12	₹61
Higher ondary	Sec-64.7%	49	12	₹109.36

Secondary Infrastructure & Fleet Summary

- Teachers Required: 110

• Girls Hostels: 0

• Cyclone Retrofits: 43
- Transit Minibuses: 12

• Transit Feeder Vans: 12

• Opex: ₹0.96 Cr/yr

Bargarh District

Western Agrarian | Vulnerability: 58/100 | GPI: 0.95 | Hill Cost Index: 1.031

BARGARH DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Western Plain | 12 CD Blocks | Walking Friction: 1.17x (Tobler) | PWD Hill Cost Index: 1.03x

Block

Base

Upgr

New

Cost

Gaisilet

70%

6

2

₹10.59

Barpali

71%

5

2

₹9.71

Sohela

71%

5

1

₹7.2

Attabira

71.7%

5

2

₹9.71

Bheden

71.7%

6

2

₹10.59

Strategic Directives

Bargarh (Western Agrarian, Vulnerability: 58/100) comprises 12 CD blocks spanning 5,837 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.95, baseline secondary access is 72.5%. The masterplan allocates 61 high school upgrades, 20 greenfield campuses, 12 student transport hubs (12 mini-buses, 12 vans), 0 dedicated girls' hostels, and 202 subject teacher recruitments. Total estimated capital outlay is Rs. 107.39 Crores, raising universal secondary coverage to 98.3% across all 12 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	82.5%	36	12	₹31.68
Upper Primary	76.5%	51	15	₹56.1
Secondary	72.5%	61	20	₹107.39
Higher ondary	Sec-	92	32	₹234.74

Secondary Infrastructure & Fleet Summary

Teachers Required: 202

Girls Hostels: 0

Cyclone Retrofits: 0

Transit Minibuses: 12

Transit Feeder Vans: 12

Opex: ₹0.96 Cr/yr

Decision Support Framework: Operations Research & Spatial Optimization

Page 9 of 37

Bhadrak District

Northern Coastal | Vulnerability: 44/100 | GPI: 0.95 | Hill Cost Index: 1.031

BHADRAK DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Coastal Plain | 7 CD Blocks | Walking Friction: 1.17x (Tobler) | PWD Hill Cost Index: 1.03x

Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Bhadrak	76.2%	3	1	₹5.45
Basudevpur	76.6%	3	1	₹5.45
Tihidi	77%	3	1	₹5.45
Bonth	78%	3	1	₹5.45
Chandabali	78%	2	1	₹4.57

Strategic Directives

Bhadrak (Northern Coastal, Vulnerability: 44/100) comprises 7 CD blocks spanning 2,505 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.95, baseline secondary access is 77.8%. The masterplan allocates 20 high school upgrades, 7 greenfield campuses, 7 student transport hubs (7 mini-buses, 7 vans), 0 dedicated girls' hostels, and 68 subject teacher recruitments. Total estimated capital outlay is Rs. 37.27 Crores, raising universal secondary coverage to 98.7% across all 7 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	87.8%	9	7	₹13.27
Upper Primary	81.8%	16	7	₹22
Secondary	77.8%	20	7	₹37.27
Higher ondary	64.8%	34	13	₹92.17

Secondary Infrastructure & Fleet Summary

Teachers Required: 68

Girls Hostels: 0

Cyclone Retrofits: 27

Transit Minibuses: 7

Transit Feeder Vans: 7

Opex: ₹0.56 Cr/yr

Boudh District

Central Tribal | Vulnerability: 72/100 | GPI: 0.91 | Hill Cost Index: 1.265

BOUDH DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Central Riverine | 3 CD Blocks | Walking Friction: 1.17x (Tobler) | PWD Hill Cost Index: 1.03x

Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Boudh	41.4%	21	14	₹70.89
Kantamal	42.8%	22	15	₹75.06
Harabhanga	43.9%	25	16	₹81.97

Strategic Directives

Boudh (Central Tribal, Vulnerability: 72/100) comprises 3 CD blocks spanning 3,098 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.91, baseline secondary access is 42.9%. The masterplan allocates 68 high school upgrades, 45 greenfield campuses, 53 student transport hubs (84 mini-buses, 47 vans), 31 dedicated girls' hostels, and 316 subject teacher recruitments. Total estimated capital outlay is Rs. 227.92 Crores, raising universal secondary coverage to 96.6% across all 3 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	64.9%	41	28	₹67.17
Upper Primary	56.9%	50	34	₹111.59
Secondary	42.9%	68	45	₹227.92
Higher ondary	31%	82	55	₹396.79

Secondary Infrastructure & Fleet Summary

Teachers Required: 316

Girls Hostels: 31

Cyclone Retrofits: 0

Transit Minibuses: 84

Transit Feeder Vans: 47

Opex: ₹5.45 Cr/yr

Decision Support Framework: Operations Research & Spatial Optimization

Page 10 of 37

Cuttack District | Central Coastal | Vulnerability: 35/100 | GPI: 0.97 | Hill Cost Index: 1.031

CUTTACK DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Delta / Urban | 14 CD Blocks | Walking Friction: 1.17x (Tobler) | PWD Hill Cost Index: 1.03x

Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Niali	78%	2	1	₹4.57
Nischintakoili	78.3%	2	1	₹4.57
Tangi-Choudwar	78.3%	2	1	₹4.57
Cuttack Sadar	78.7%	2	1	₹4.57
Banki-Dampada	79%	2	1	₹4.57

Strategic Directives

Cuttack (Central Coastal, Vulnerability: 35/100) comprises 14 CD blocks spanning 3,932 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.97, baseline secondary access is 80.3%. The masterplan allocates 26 high school upgrades, 14 greenfield campuses, 14 student transport hubs (14 mini-buses, 14 vans), 0 dedicated girls' hostels, and 108 subject teacher recruitments. Total estimated capital outlay is Rs. 62.24 Crores, raising universal secondary coverage to 98.8% across all 14 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr	
Primary	90.4%	14	14	₹24.78	
Upper Primary	84.3%	19	14	₹36.68	
Secondary	80.3%	26	14	₹62.24	
Higher ondry	Sec-	67.3%	49	14	₹117.46

Secondary Infrastructure & Fleet Summary

Teachers Required: 108

Girls Hostels: 0

Cyclone Retrofits: 40

Transit Minibuses: 14

Transit Feeder Vans: 14

Opex: ₹1.12 Cr/yr

Deogarh District | Northern Tribal | Vulnerability: 65/100 | GPI: 0.92 | Hill Cost Index: 1.265

DEOGARH DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Northern Hilly | 3 CD Blocks | Walking Friction: 2.47x (Tobler) | PWD Hill Cost Index: 1.26x

Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Tileibani	43.9%	21	14	₹70.89
Barkote	44.2%	18	12	₹60.59
Reamal	46%	19	12	₹61.97

Strategic Directives

Deogarh (Northern Tribal, Vulnerability: 65/100) comprises 3 CD blocks spanning 2,782 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.92, baseline secondary access is 44.8%. The masterplan allocates 58 high school upgrades, 38 greenfield campuses, 46 student transport hubs (73 mini-buses, 40 vans), 27 dedicated girls' hostels, and 268 subject teacher recruitments. Total estimated capital outlay is Rs. 193.45 Crores, raising universal secondary coverage to 96.7% across all 3 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr	
Primary	66.8%	35	22	₹54.49	
Upper Primary	58.7%	44	29	₹96.11	
Secondary	44.8%	58	38	₹193.45	
Higher ondry	Sec-	32.8%	71	47	₹340.62

Secondary Infrastructure & Fleet Summary

Teachers Required: 268

Girls Hostels: 27

Cyclone Retrofits: 0

Transit Minibuses: 73

Transit Feeder Vans: 40

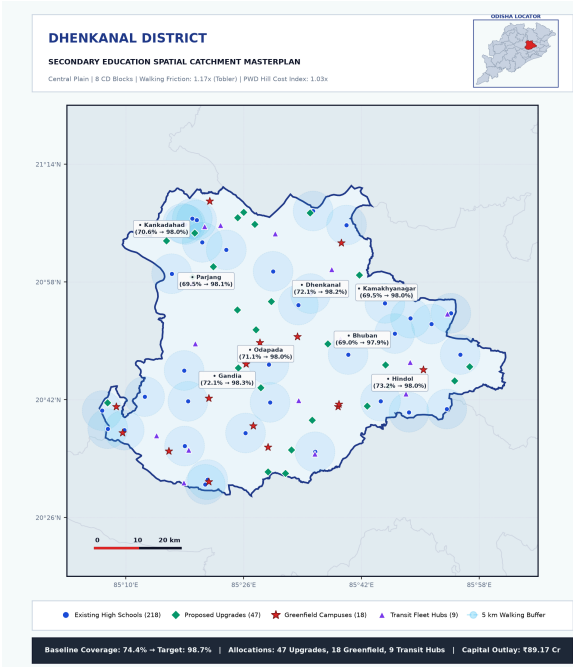
Opex: ₹4.71 Cr/yr

Decision Support Framework: Operations Research & Spatial Optimization

Page 11 of 37

Dhenkanal District

Central Tribal | Vulnerability: 55/100 | GPI: 0.94 | Hill Cost Index: 1.265



Strategic Directives

Dhenkanal (Central Tribal, Vulnerability: 55/100) comprises 8 CD blocks spanning 4,452 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.94, baseline secondary access is 49.1%. The masterplan allocates 82 high school upgrades, 54 greenfield campuses, 62 student transport hubs (96 mini-buses, 54 vans), 34 dedicated girls' hostels, and 380 subject teacher recruitments. Total estimated capital outlay is Rs. 273.45 Crores, raising universal secondary coverage to 97.0% across all 8 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	71.1%	45	28	₹69.9
Upper Primary	63.1%	57	37	₹123.56
Secondary	49.1%	82	54	₹273.45
Higher ondary	37.1%	102	67	₹486.57

Secondary Infrastructure & Fleet Summary

- Teachers Required: 380

Girls Hostels: 34

Cyclone Retrofits: 0
- Transit Minibuses: 96

Transit Feeder Vans: 54

Opex: ₹6.24 Cr/yr

Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Bhuban	47.7%	11	7	₹35.84
Kamakhyanager	48%	9	6	₹30.3
Parjang	48%	12	8	₹40.29
Kankadahad	48.8%	10	7	₹34.76
Odapada	49.1%	12	8	₹40.29

GAJAPATI DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Tribal / Hilly | 7 CD Blocks | Walking Friction: 1.17x (Tobler) | PWD Hill Cost Index: 1.05x

Strategic Directives

Gajapati (Southern Tribal, Vulnerability: 88/100) comprises 7 CD blocks spanning 3,850 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.88, baseline secondary access is 37.5%. The masterplan allocates 94 high school upgrades, 62 greenfield campuses, 73 student transport hubs (114 mini-buses, 62 vans), 41 dedicated girls' hostels, and 436 subject teacher recruitments. Total estimated capital outlay is Rs. 314.33 Crores, raising universal secondary coverage to 96.2% across all 7 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	59.6%	60	39	₹95.5
Upper Primary	51.6%	72	47	₹156.22
Secondary	37.5%	94	62	₹314.33
Higher onduary	Sec-30.2%	104	70	₹505.14

Secondary Infrastructure & Fleet Summary

Teachers Required: 436

Girls Hostels: 41

Cyclone Retrofits: 0

Transit Minibuses: 114

Transit Feeder Vans: 62

Opex: ₹7.35 Cr/yr

Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
R.Udayagiri	35.5%	13	8	₹41.67
Parlakhemundi	36.2%	12	8	₹40.29
Kashinagar	37.2%	15	10	₹50.3
Mohana	37.2%	12	8	₹40.59
Nuagada	37.9%	14	9	₹46.13

Strategic Directives

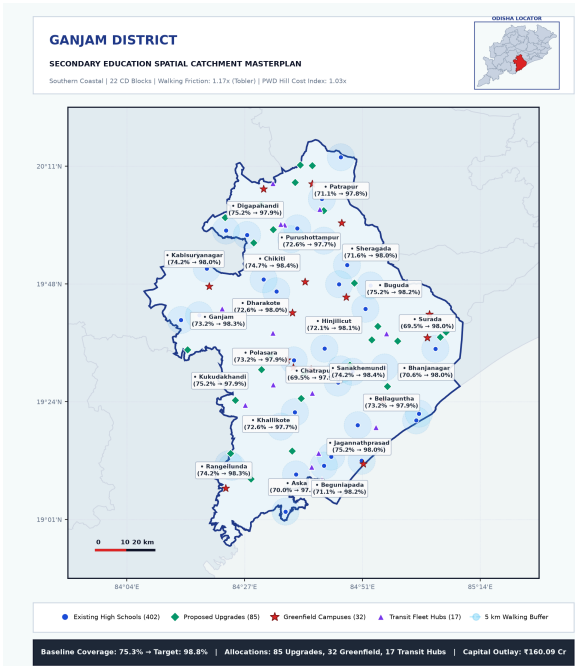
Ganjam (Southern Coastal, Vulnerability: 59/100) comprises 22 CD blocks spanning 8,206 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.93, baseline secondary access is 72.2%. The masterplan allocates 87 high school upgrades, 22 greenfield campuses, 22 student transport hubs (22 mini-buses, 22 vans), 0 dedicated girls' hostels, and 262 subject teacher recruitments. Total estimated capital outlay is Rs. 138.35 Crores, raising universal secondary coverage to 98.3% across all 22 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	82.3%	51	22	₹51.63
Upper Primary	76.2%	71	22	₹80.95
Secondary	72.2%	87	22	₹138.35
Higher Sec- ondary	59.2%	131	44	₹330.78

Secondary Infrastructure & Fleet Summary

- Teachers Required: **262**
- Girls Hostels: **0**
- Cyclone Retrofits: **109**
- Transit Minibuses: **22**
- Transit Feeder Vans: **22**
- Opex: **₹1.76 Cr/yr**

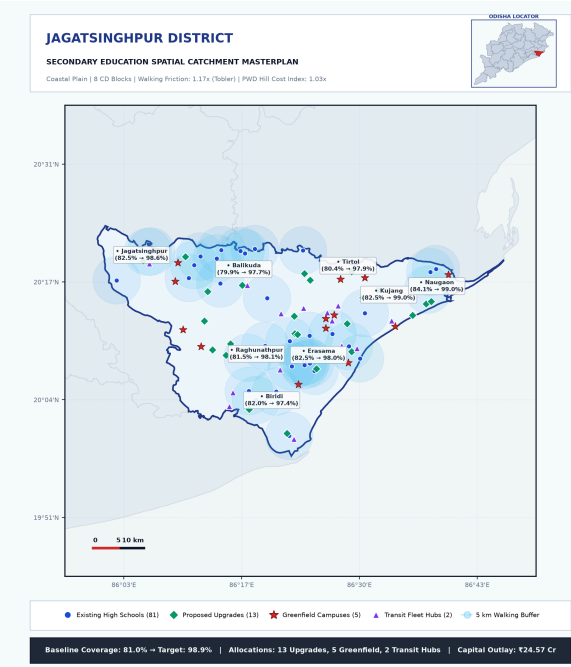


Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Chatrapur	69.6%	5	1	₹7.2
Surada	69.6%	4	1	₹6.33
Aska	70%	5	1	₹7.2
Bhanjanagar	70.3%	5	1	₹7.2
Patrapur	70.7%	3	1	₹5.45

Jagatsinghpur District

Central Coastal | Vulnerability: 38/100 | GPI: 0.98 | Hill Cost Index: 1.031



Strategic Directives

Jagatsinghpur (Central Coastal, Vulnerability: 38/100) comprises 8 CD blocks spanning 1,668 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.98, baseline secondary access is 79.6%. The masterplan allocates 8 high school upgrades, 8 greenfield campuses, 8 student transport hubs (8 mini-buses, 8 vans), 0 dedicated girls' hostels, and 48 subject teacher recruitments. Total estimated capital outlay is Rs. 29.6 Crores, raising universal secondary coverage to 98.8% across all 8 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	89.4%	8	8	₹14.16
Upper Primary	83.3%	8	8	₹19.36
Secondary	79.6%	8	8	₹29.6
Higher ondary	Sec- 66.5%	19	8	₹55.96

Secondary Infrastructure & Fleet Summary

- Teachers Required: 48

Girls Hostels: 0

Cyclone Retrofits: 16
- Transit Minibuses: 8

Transit Feeder Vans: 8

Opex: ₹0.64 Cr/yr

Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Balikuda	77.3%	1	1	₹3.7
Tirtol	77.7%	1	1	₹3.7
Raghunathpur	78.3%	1	1	₹3.7
Biridi	78.7%	1	1	₹3.7
Erasama	79%	1	1	₹3.7

Jajpur District | Northern Coastal | Vulnerability: 45/100 | GPI: 0.95 | Hill Cost Index: 1.031

JAJPUR DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Coastal Industrial | 10 CD Blocks | Walking Friction: 1.17x (Tobler) | PWD Hill Cost Index: 1.03x

Existing High Schools (141)

Proposed Upgrades (26)

Greenfield Campuses (10)

Transit Fleet Hubs (5)

5 km Walking Buffer

Baseline Coverage: 78.0% → Target: 98.9% | Allocations: 26 Upgrades, 10 Greenfield, 5 Transit Hubs | Capital Outlay: ₹49.44 Cr

Strategic Directives

Jajpur (Northern Coastal, Vulnerability: 45/100) comprises 10 CD blocks spanning 2,899 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.95, baseline secondary access is 77.0%. The masterplan allocates 24 high school upgrades, 10 greenfield campuses, 10 student transport hubs (10 mini-buses, 10 vans), 0 dedicated girls' hostels, and 88 subject teacher recruitments. Total estimated capital outlay is Rs. 49.22 Crores, raising universal secondary coverage to 98.6% across all 10 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	86.9%	10	10	₹17.7
Upper Primary	81.1%	20	10	₹29.8
Secondary	77%	24	10	₹49.22
Higher ondry	64.1%	38	10	₹87.62

Secondary Infrastructure & Fleet Summary

Teachers Required: 88

Transit Minibuses: 10

Girls Hostels: 0

Transit Feeder Vans: 10

Cyclone Retrofits: 34

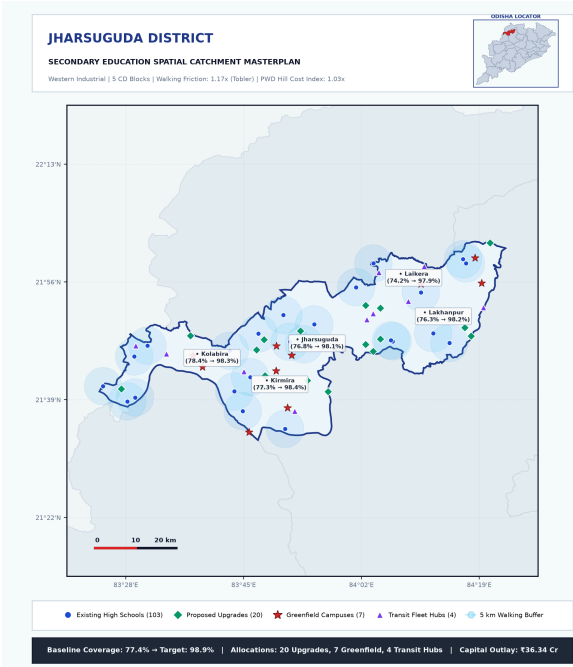
Opex: ₹0.8 Cr/yr

Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Danagadi	74.5%	3	1	₹5.45
Barchana	75.5%	2	1	₹4.57
Dasarathpur	75.5%	3	1	₹5.45
Bari	76.2%	2	1	₹4.57
Sukinda	76.6%	3	1	₹5.45

Jharsuguda District

Western Industrial | Vulnerability: 52/100 | GPI: 0.93 | Hill Cost Index: 1.031



Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Laikera	72.8%	4	1	₹6.33
Lakhampur	74.2%	4	1	₹6.33
Jharsuguda	74.5%	4	1	₹6.33
Kirmira	74.8%	3	1	₹5.45
Kolabira	75.5%	4	1	₹6.33

Strategic Directives

Jharsuguda (Western Industrial, Vulnerability: 52/100) comprises 5 CD blocks spanning 2,081 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.93, baseline secondary access is 74.7%. The masterplan allocates 19 high school upgrades, 5 greenfield campuses, 5 student transport hubs (5 mini-buses, 5 vans), 0 dedicated girls' hostels, and 58 subject teacher recruitments. Total estimated capital outlay is Rs. 30.77 Crores, raising universal secondary coverage to 98.5% across all 5 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	84.8%	11	5	₹11.48
Upper Primary	78.6%	18	5	₹19.46
Secondary	74.7%	19	5	₹30.77
Higher onduary	61.8%	32	10	₹77.94

Secondary Infrastructure & Fleet Summary

- Teachers Required: 58

Girls Hostels: 0

Cyclone Retrofits: 0
- Transit Minibuses: 5

Transit Feeder Vans: 5

Opex: ₹0.4 Cr/yr

Kalahandi District

Western Tribal | Vulnerability: 84/100 | GPI: 0.9 | Hill Cost Index: 1.265

KALAHANDI DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Drought-prone / Tribal | 13 CD Blocks | Walking Friction: 1.17x (Tobler) | PWD Hill Cost Index: 1.03x

ODISHA LOCATION

Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Kalampur	36.9%	11	7	₹35.84
Golamunda	37.2%	12	8	₹40.29
Kesinga	37.2%	13	9	₹44.76
Jaipatna	37.5%	12	8	₹40.29
Koksara	37.5%	11	8	₹39.22

Strategic Directives

Kalahandi (Western Tribal, Vulnerability: 84/100) comprises 13 CD blocks spanning 7,920 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.9, baseline secondary access is 38.5%. The masterplan allocates 150 high school upgrades, 101 greenfield campuses, 115 student transport hubs (178 mini-buses, 102 vans), 63 dedicated girls' hostels, and 704 subject teacher recruitments. Total estimated capital outlay is Rs. 507.53 Crores, raising universal secondary coverage to 96.3% across all 13 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	60.6%	91	61	₹148.02
Upper Primary	52.6%	115	74	₹247.38
Secondary	38.5%	150	101	₹507.53
Higher Sec-ondary	30.2%	171	114	₹823.26

Secondary Infrastructure & Fleet Summary

Teachers Required: 704

Girls Hostels: 63

Cyclone Retrofits: 0

Transit Minibuses: 178

Transit Feeder Vans: 102

Opex: ₹11.62 Cr/yr

Decision Support Framework: Operations Research & Spatial Optimization

Page 18 of 37

Kandhamal District

Central Tribal | Vulnerability: 92/100 | GPI: 0.87 | Hill Cost Index: 1.265

KANDHAMAL DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Tribal / Hilly | 12 CD Blocks | Walking Friction: 2.47x (Tobler) | PWD Hill Cost Index: 1.26x

Existing High Schools (272)

Proposed Upgrades (151)

Greenfield Campuses (101)

Transit Fleet Hubs (118)

5 km Walking Buffer

Baseline Coverage: 40.8% → Target: 97.0% | Allocations: 151 Upgrades, 101 Greenfield, 118 Transit Hubs | Capital Outlay: ₹509.51 Cr

Strategic Directives

Kandhamal (Central Tribal, Vulnerability: 92/100) comprises 12 CD blocks spanning 8,021 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.87, baseline secondary access is 36.1%. The masterplan allocates 159 high school upgrades, 105 greenfield campuses, 124 student transport hubs (194 mini-buses, 106 vans), 70 dedicated girls' hostels, and 738 subject teacher recruitments. Total estimated capital outlay is Rs. 532.27 Crores, raising universal secondary coverage to 96.2% across all 12 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	58.1%	102	67	₹163.64
Upper Primary	50.1%	123	82	₹270.51
Secondary	36.1%	159	105	₹532.27
Higher ondry	Sec- 30.2%	172	114	₹827.02

Secondary Infrastructure & Fleet Summary

Teachers Required: 738

Girls Hostels: 70

Cyclone Retrofits: 0

Transit Minibuses: 194

Transit Feeder Vans: 106

Opex: ₹12.5 Cr/yr

Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Daringbadi	34%	15	10	₹50.3
Khajuripada	34.8%	12	8	₹40.29
Chakapada	35.1%	13	9	₹44.76
Phulbani	35.1%	15	10	₹50.6
G.Udayagiri	35.5%	15	10	₹50.6

Decision Support Framework: Operations Research & Spatial Optimization

Page 19 of 37

Kendrapara District

Central Coastal | Vulnerability: 40/100 | GPI: 0.97 | Hill Cost Index: 1.031

KENDRAPARA DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Coastal Wetland | 9 CD Blocks | Walking Friction: 1.17x (Tobler) | PWD Hill Cost Index: 1.03x

Existing High Schools (129)

Proposed Upgrades (25)

Greenfield Campuses (9)

Transit Fleet Hubs (5)

5 km Walking Buffer

Baseline Coverage: 76.8% → Target: 98.8% | Allocations: 25 Upgrades, 9 Greenfield, 5 Transit Hubs | Capital Outlay: ₹46.05 Cr

Block

Base

Upgr

New

Cost

Marshaghai

77%

2

1

₹4.57

Rajakanika

77.3%

2

1

₹4.57

Mahakalapada

78%

2

1

₹4.57

Pattamundai

78.3%

2

1

₹4.57

Aul

79.4%

2

1

₹4.57

Strategic Directives

Kendrapara (Central Coastal, Vulnerability: 40/100) comprises 9 CD blocks spanning 2,644 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.97, baseline secondary access is 79.2%. The masterplan allocates 18 high school upgrades, 9 greenfield campuses, 9 student transport hubs (9 mini-buses, 9 vans), 0 dedicated girls' hostels, and 72 subject teacher recruitments. Total estimated capital outlay is Rs. 41.13 Crores, raising universal secondary coverage to 98.8% across all 9 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	89.1%	9	9	₹15.93
Upper Primary	83.2%	14	9	₹24.58
Secondary	79.2%	18	9	₹41.13
Higher onduary	Sec- 66.1%	31	9	₹74.89

Secondary Infrastructure & Fleet Summary

Teachers Required: 72

Girls Hostels: 0

Cyclone Retrofits: 27

Transit Minibuses: 9

Transit Feeder Vans: 9

Opex: ₹0.72 Cr/yr

Decision Support Framework: Operations Research & Spatial Optimization

Page 20 of 37

Kendujhar District

Northern Tribal | Vulnerability: 78/100 | GPI: 0.91 | Hill Cost Index: 1.265

KENDUJHAR DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Tribal / Mining | 13 CD Blocks | Walking Friction: 2.47x (Tobler) | PWD Hill Cost Index: 1.26x

Existing High Schools (272)

Proposed Upgrades (142)

Greenfield Campuses (95)

Transit Fleet Hubs (111)

5 km Walking Buffer

Baseline Coverage: 44.4% – Target: 97.2% | Allocations: 142 Upgrades, 95 Greenfield, 111 Transit Hubs | Capital Outlay: ₹479.21 Cr

Strategic Directives

Kendujhar (Northern Tribal, Vulnerability: 78/100) comprises 13 CD blocks spanning 8,303 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.91, baseline secondary access is 41.1%. The masterplan allocates 151 high school upgrades, 98 greenfield campuses, 114 student transport hubs (178 mini-buses, 101 vans), 64 dedicated girls' hostels, and 694 subject teacher recruitments. Total estimated capital outlay is Rs. 499.03 Crores, raising universal secondary coverage to 96.5% across all 13 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr	
Primary	63.1%	91	59	₹144.67	
Upper Primary	55.1%	114	74	₹246.28	
Secondary	41.1%	151	98	₹499.03	
Higher ondary	Sec- ondary	30.4%	177	119	₹856.73

Secondary Infrastructure & Fleet Summary

Teachers Required: 694

Girls Hostels: 64

Cyclone Retrofits: 0

Transit Minibuses: 178

Transit Feeder Vans: 101

Opex: ₹11.59 Cr/yr

Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Anandapur	39%	13	8	₹41.67
Champua	39%	13	8	₹41.67
Telkoi	39%	10	7	₹34.76
Keonjhar	39.3%	13	9	₹44.76
Ghasipura	39.6%	10	7	₹34.76

Decision Support Framework: Operations Research & Spatial Optimization

Page 21 of 37

Strategic Directives

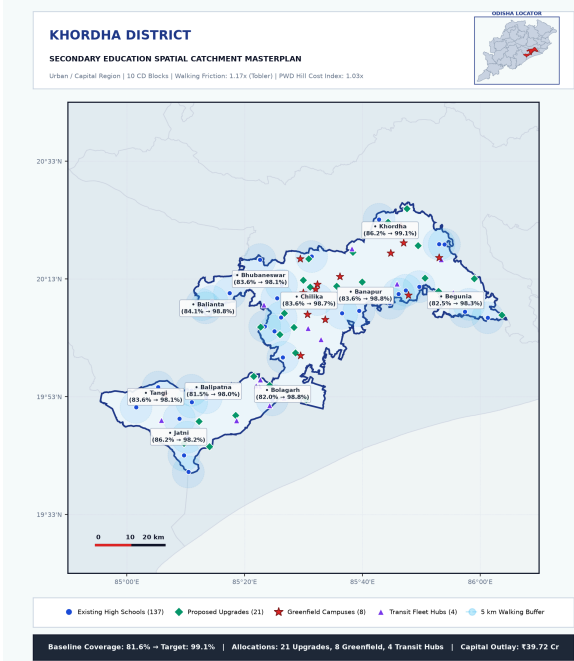
Khordha (Central Coastal, Vulnerability: 30/100) comprises 10 CD blocks spanning 2,813 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.98, baseline secondary access is 81.7%. The masterplan allocates 15 high school upgrades, 10 greenfield campuses, 10 student transport hubs (10 mini-buses, 10 vans), 0 dedicated girls' hostels, and 70 subject teacher recruitments. Total estimated capital outlay is Rs. 41.35 Crores, raising universal secondary coverage to 98.9% across all 10 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	91.8%	10	10	₹17.7
Upper Primary	85.7%	10	10	₹24.2
Secondary	81.7%	15	10	₹41.35
Higher Sec-ondary	68.6%	31	10	₹78.94

Secondary Infrastructure & Fleet Summary

- Teachers Required: **70**
- Girls Hostels: **0**
- Cyclone Retrofits: **0**
- Transit Minibuses: **10**
- Transit Feeder Vans: **10**
- Opex: **₹0.8 Cr/yr**

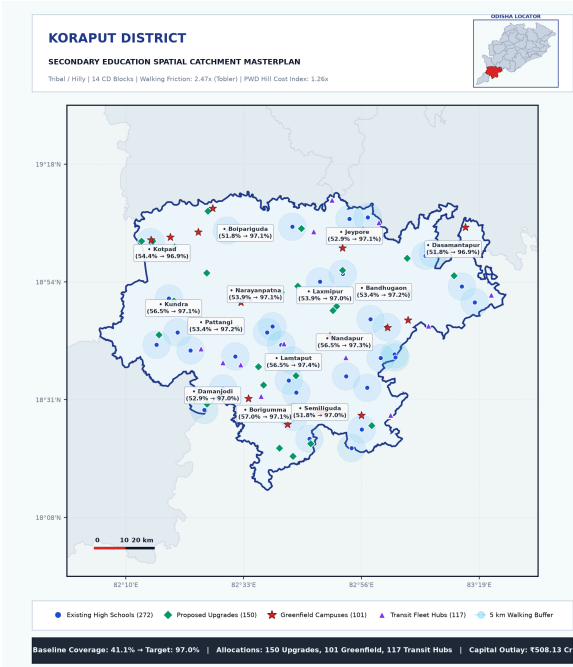


Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Balipatna	80.5%	2	1	₹4.57
Bolagarh	80.8%	2	1	₹4.57
Banapur	81.2%	1	1	₹3.7
Baliana	81.5%	1	1	₹3.7
Bhubaneswar	81.5%	2	1	₹4.57

Koraput District

Southern Tribal | Vulnerability: 91/100 | GPI: 0.86 | Hill Cost Index: 1.265



Strategic Directives

Koraput (Southern Tribal, Vulnerability: 91/100) comprises 14 CD blocks spanning 8,807 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.86, baseline secondary access is 36.2%. The masterplan allocates 156 high school upgrades, 103 greenfield campuses, 117 student transport hubs (179 mini-buses, 103 vans), 62 dedicated girls' hostels, and 724 subject teacher recruitments. Total estimated capital outlay is Rs. 520.75 Crores, raising universal secondary coverage to 96.2% across all 14 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	58.1%	99	64	₹157.09
Upper Primary	50.1%	118	77	₹256.13
Secondary	36.2%	156	103	₹520.75
Higher Sec-ondary	30.2%	171	112	₹813.96

Secondary Infrastructure & Fleet Summary

- Teachers Required: 724

Girls Hostels: 62

Cyclone Retrofits: 0
- Transit Minibuses: 179

Transit Feeder Vans: 103

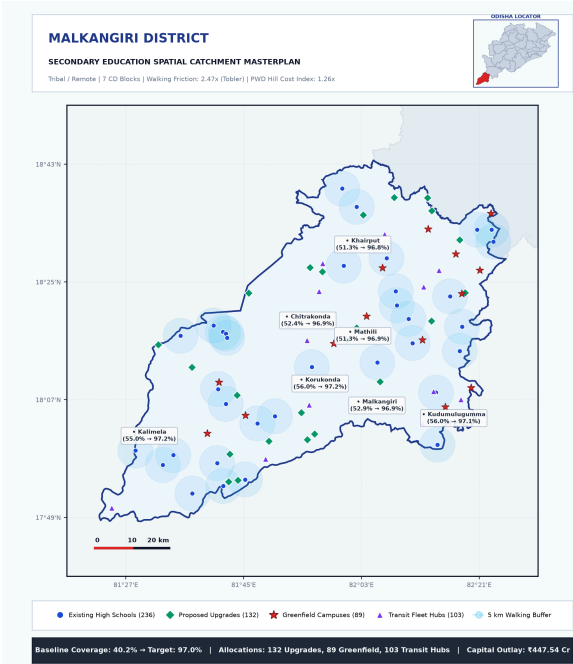
Opex: ₹11.7 Cr/yr

Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Bandhugaon	34.4%	11	7	₹35.84
Dasamantapur	34.4%	11	7	₹35.84
Narayanpatna	34.4%	12	8	₹40.29
Borigumma	35.1%	12	8	₹40.29
Jeypore	35.1%	12	8	₹40.29

Malkangiri District

Southern Tribal | Vulnerability: 95/100 | GPI: 0.85 | Hill Cost Index: 1.265



Strategic Directives

Malkangiri (Southern Tribal, Vulnerability: 95/100) comprises 7 CD blocks spanning 5,791 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.85, baseline secondary access is 35.3%. The masterplan allocates 127 high school upgrades, 86 greenfield campuses, 99 student transport hubs (155 mini-buses, 85 vans), 56 dedicated girls' hostels, and 598 subject teacher recruitments. Total estimated capital outlay is Rs. 431.69 Crores, raising universal secondary coverage to 96.1% across all 7 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	57.4%	82	55	₹133.32
Upper Primary	49.4%	100	65	₹216.75
Secondary	35.3%	127	86	₹431.69
Higher ondary	Sec-30.1%	139	91	₹661.6

Secondary Infrastructure & Fleet Summary

- Teachers Required: 598

Girls Hostels: 56

Cyclone Retrofits: 0
- Transit Minibuses: 155

Transit Feeder Vans: 85

Opex: ₹10.01 Cr/yr

Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Khairput	33.4%	18	12	₹60.59
Podia	33.7%	17	12	₹59.52
Chitrakonda	34.4%	17	12	₹59.22
Mathili	34.8%	18	12	₹60.59
Kalimela	36.2%	21	14	₹70.59

Mayurbhanj District

Northern Tribal | Vulnerability: 82/100 | GPI: 0.89 | Hill Cost Index: 1.265

MAYURBHANJ DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Tribal / Forest | 26 CD Blocks | Walking Friction: 2.47x (Tobler) | PWD Hill Cost Index: 1.26x

Existing High Schools (272)

Proposed Upgrades (145)

Greenfield Campuses (97)

Transit Fleet Hubs (113)

5 km Walking Buffer

Baseline Coverage: 43.2% – Target: 97.1% | Allocations: 145 Upgrades, 97 Greenfield, 113 Transit Hubs | Capital Outlay: ₹489.20 Cr

Strategic Directives

Mayurbhanj (Northern Tribal, Vulnerability: 82/100) comprises 26 CD blocks spanning 10,418 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.89, baseline secondary access is 39.5%. The masterplan allocates 140 high school upgrades, 89 greenfield campuses, 110 student transport hubs (168 mini-buses, 84 vans), 58 dedicated girls' hostels, and 636 subject teacher recruitments. Total estimated capital outlay is Rs. 458.32 Crores, raising universal secondary coverage to 96.4% across all 26 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	61.5%	87	52	₹131.31
Upper Primary	53.7%	104	65	₹219.66
Secondary	39.5%	140	89	₹458.32
Higher ondry	Sec-30.5%	163	107	₹777.05

Secondary Infrastructure & Fleet Summary

Teachers Required: 636

Transit Minibuses: 168

Girls Hostels: 58

Transit Feeder Vans: 84

Cyclone Retrofits: 0

Opex: ₹10.59 Cr/yr

Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Badasahi	37.5%	5	3	₹15.84
Barsahi	37.5%	5	3	₹15.84
Gopabandhunagar	37.5%	6	4	₹20.3
Jashipur	37.5%	5	3	₹15.84
Bijatala	37.9%	5	3	₹15.84

Nabarangpur District

Southern Tribal | Vulnerability: 89/100 | GPI: 0.87 | Hill Cost Index: 1.265

NABARANGPUR DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Tribal / Plateau | 10 CD Blocks | Walking Friction: 2.47x (Tobler) | PWD Hill Cost Index: 1.26x

Existing High Schools (215)

Proposed Upgrades (116)

Greenfield Campuses (78)

Transit Fleet Hubs (91)

5 km Walking Buffer

Baseline Coverage: 42.6% → Target: 97.1% | Allocations: 116 Upgrades, 78 Greenfield, 91 Transit Hubs | Capital Outlay: ₹392.77 Cr

Strategic Directives

Nabarangpur (Southern Tribal, Vulnerability: 89/100) comprises 10 CD blocks spanning 5,291 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.87, baseline secondary access is 37.6%. The masterplan allocates 118 high school upgrades, 77 greenfield campuses, 93 student transport hubs (144 mini-buses, 81 vans), 51 dedicated girls' hostels, and 544 subject teacher recruitments. Total estimated capital outlay is Rs. 392.45 Crores, raising universal secondary coverage to 96.3% across all 10 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr	
Primary	59.6%	76	49	₹120.2	
Upper Primary	51.6%	88	59	₹194.44	
Secondary	37.6%	118	77	₹392.45	
Higher ondary	Sec- ondary	30.2%	132	86	₹627.94

Secondary Infrastructure & Fleet Summary

Teachers Required: 544

Girls Hostels: 51

Cyclone Retrofits: 0

Transit Minibuses: 144

Transit Feeder Vans: 81

Opex: ₹9.36 Cr/yr

Priority Block Allocations (Top 5 Vulnerable)

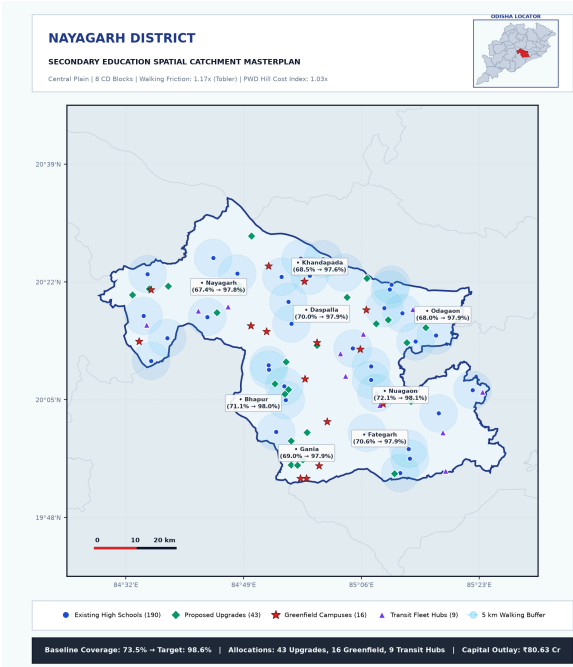
Block	Base	Upgr	New	Cost
Kosagumuda	35.5%	12	8	₹40.29
Chandahandi	36.2%	14	9	₹46.13
Tentulikhunti	36.9%	11	7	₹35.84
Nabarangpur	37.5%	14	9	₹46.13
Raighar	37.5%	11	7	₹36.14

Decision Support Framework: Operations Research & Spatial Optimization

Page 26 of 37

Nayagarh District

Central Coastal | Vulnerability: 48/100 | GPI: 0.94 | Hill Cost Index: 1.031



Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Nuagaon	74.2%	5	2	₹9.71
Ranpur	74.5%	5	2	₹9.71
Nayagarh	74.8%	5	2	₹9.71
Khandapada	75.2%	5	2	₹9.71
Dasapalla	75.9%	5	1	₹7.2

Strategic Directives

Nayagarh (Central Coastal, Vulnerability: 48/100) comprises 8 CD blocks spanning 3,890 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.94, baseline secondary access is 75.8%. The masterplan allocates 37 high school upgrades, 12 greenfield campuses, 8 student transport hubs (8 mini-buses, 8 vans), 0 dedicated girls' hostels, and 122 subject teacher recruitments. Total estimated capital outlay is Rs. 65.03 Crores, raising universal secondary coverage to 98.6% across all 8 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	85.9%	21	8	₹19.83
Upper Primary	79.7%	29	8	₹31.25
Secondary	75.8%	37	12	₹65.03
Higher ondary	62.8%	57	21	₹149.93

Secondary Infrastructure & Fleet Summary

- Teachers Required: 122

Girls Hostels: 0

Cyclone Retrofits: 0
- Transit Minibuses: 8

Transit Feeder Vans: 8

Opex: ₹0.64 Cr/yr

Strategic Directives

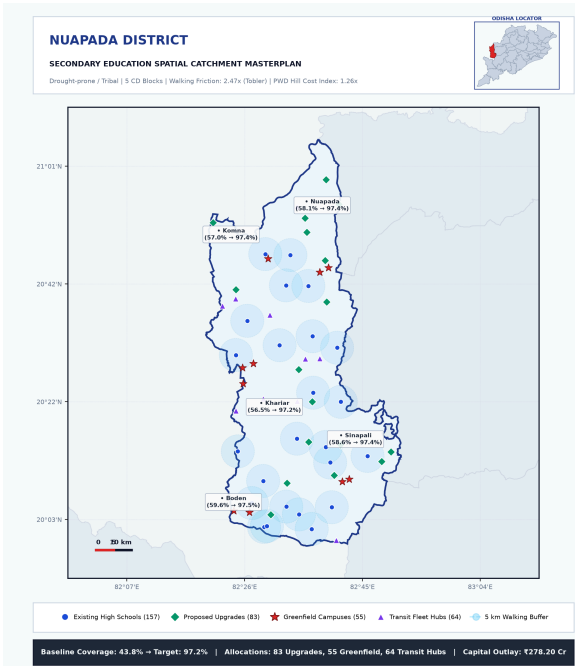
Nuapada (Western Tribal, Vulnerability: 86/100) comprises 5 CD blocks spanning 3,852 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.9, baseline secondary access is 61.2%. The masterplan allocates 54 high school upgrades, 20 greenfield campuses, 10 student transport hubs (10 mini-buses, 5 vans), 0 dedicated girls' hostels, and 188 subject teacher recruitments. Total estimated capital outlay is Rs. 100.62 Crores, raising universal secondary coverage to 97.6% across all 5 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	71.3%	41	14	₹35.26
Upper Primary	65.3%	49	17	₹57.63
Secondary	61.2%	54	20	₹100.62
Higher Sec- ondary	48.2%	75	27	₹196.51

Secondary Infrastructure & Fleet Summary

- Teachers Required: **188**
- Girls Hostels: **0**
- Cyclone Retrofits: **0**
- Transit Minibuses: **10**
- Transit Feeder Vans: **5**
- Opex: **₹0.65 Cr/yr**



Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Khariar	60.2%	12	4	₹21.18
Komna	60.5%	10	4	₹19.42
Nuapada	61.2%	12	4	₹21.18
Sinapali	61.5%	10	4	₹19.42
Boden	62.2%	10	4	₹19.42

Strategic Directives

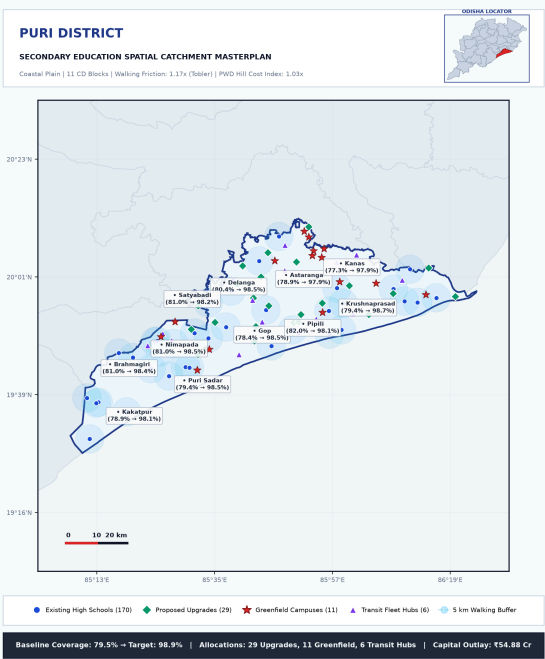
Puri (Central Coastal, Vulnerability: 36/100) comprises 11 CD blocks spanning 3,479 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.97, baseline secondary access is 80.1%. The masterplan allocates 24 high school upgrades, 11 greenfield campuses, 11 student transport hubs (11 mini-buses, 11 vans), 0 dedicated girls' hostels, and 92 subject teacher recruitments. Total estimated capital outlay is Rs. 52.03 Crores, raising universal secondary coverage to 98.8% across all 11 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	90%	11	11	₹19.47
Upper Primary	84%	21	11	₹32.22
Secondary	80.1%	24	11	₹52.03
Higher Sec- ondary	67%	44	11	₹99.11

Secondary Infrastructure & Fleet Summary

- Teachers Required: **92**
- Girls Hostels: **0**
- Cyclone Retrofits: **35**
- Transit Minibuses: **11**
- Transit Feeder Vans: **11**
- Opex: **₹0.88 Cr/yr**



Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Kakatpur	78%	3	1	₹4.45
Gop	78.7%	2	1	₹4.57
Astaranga	79%	2	1	₹4.57
Kanas	79%	2	1	₹4.57
Krushnaprasad	79.4%	2	1	₹4.57

RAYAGADA DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Tribal / Hilly | 11 CD Blocks | Walking Friction: 2.47x (Tobler) | PWD Hill Cost Index: 1.26x

Existing High Schools (272)

Proposed Upgrades (149)

Greenfield Campuses (100)

Transit Fleet Hubs (116)

5 km Walking Buffer

Baseline Coverage: 41.7% → Target: 97.1% | Allocations: 149 Upgrades, 100 Greenfield, 116 Transit Hubs | Capital Outlay: ₹503.66 Cr

Strategic Directives

Rayagada (Southern Tribal, Vulnerability: 90/100) comprises 11 CD blocks spanning 7,073 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.88, baseline secondary access is 36.6%. The masterplan allocates 157 high school upgrades, 103 greenfield campuses, 121 student transport hubs (190 mini-buses, 102 vans), 69 dedicated girls' hostels, and 726 subject teacher recruitments. Total estimated capital outlay is Rs. 523.02 Crores, raising universal secondary coverage to 96.2% across all 11 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	58.6%	99	67	₹161.74
Upper Primary	50.6%	119	79	₹261.08
Secondary	36.6%	157	103	₹523.02
Higher ondary	30.2%	174	114	₹828.25

Secondary Infrastructure & Fleet Summary

Teachers Required: 726

Transit Minibuses: 190

Girls Hostels: 69

Transit Feeder Vans: 102

Cyclone Retrofits: 0

Opex: ₹12.19 Cr/yr

Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Gudari	35.5%	12	8	₹40.59
Chandrapur	35.8%	15	10	₹50.3
Gunupur	35.8%	16	10	₹51.67
Padmapur	35.8%	14	9	₹46.13
Kalyansingpur	36.2%	16	10	₹51.67

Sambalpur District

Western Agrarian | Vulnerability: 62/100 | GPI: 0.93 | Hill Cost Index: 1.031

SAMBALPUR DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Western Plateau | 9 CD Blocks | Walking Friction: 1.17x (Tobler) | PWD Hill Cost Index: 1.03x

ODISHA LOCATION

Block

Base

Upgr

New

Cost

Bamra

68.5%

10

3

₹16.91

Jamankira

68.5%

9

3

₹15.74

Maneswar

69.2%

8

3

₹14.86

Dhankauda

69.6%

9

3

₹15.74

Kuchinda

69.6%

8

3

₹14.86

Strategic Directives

Sambalpur (Western Agrarian, Vulnerability: 62/100) comprises 9 CD blocks spanning 6,624 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.93, baseline secondary access is 70.2%. The masterplan allocates 75 high school upgrades, 26 greenfield campuses, 10 student transport hubs (10 mini-buses, 9 vans), 0 dedicated girls' hostels, and 254 subject teacher recruitments. Total estimated capital outlay is Rs. 134.15 Crores, raising universal secondary coverage to 98.2% across all 9 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	80.2%	48	16	₹41.13
Upper Primary	74.2%	62	22	₹73.26
Secondary	70.2%	75	26	₹134.15
Higher ondary	57.3%	108	38	₹278.82

Secondary Infrastructure & Fleet Summary

Teachers Required: 254

Transit Minibuses: 10

Girls Hostels: 0

Transit Feeder Vans: 9

Cyclone Retrofits: 0

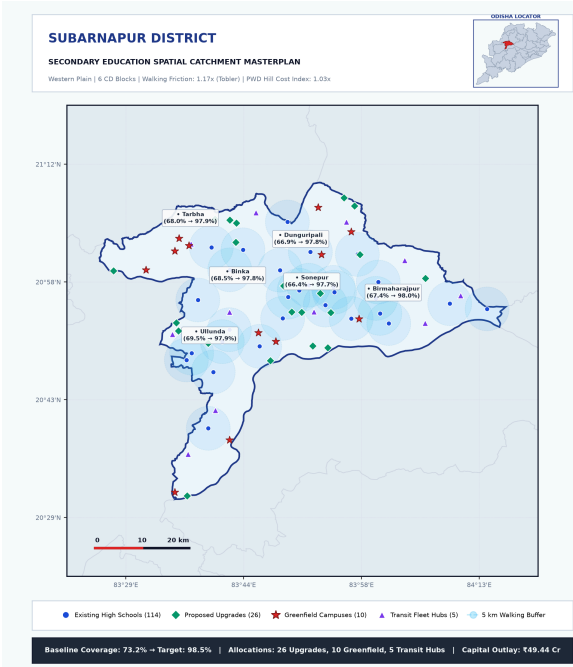
Opex: ₹0.77 Cr/yr

Decision Support Framework: Operations Research & Spatial Optimization

Page 31 of 37

Subarnapur District

Western Agrarian | Vulnerability: 60/100 | GPI: 0.94 | Hill Cost Index: 1.031



Strategic Directives

Subarnapur (Western Agrarian, Vulnerability: 60/100) comprises 6 CD blocks spanning 2,337 sq.km. With a terrain walking friction factor of 1.17x (Tobler index), a PWD Hill Cost Index of 1.031x, and a Gender Parity Index of 0.94, baseline secondary access is 70.9%. The masterplan allocates 25 high school upgrades, 7 greenfield campuses, 6 student transport hubs (6 mini-buses, 6 vans), 0 dedicated girls' hostels, and 78 subject teacher recruitments. Total estimated capital outlay is Rs. 41.36 Crores, raising universal secondary coverage to 98.2% across all 6 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	80.9%	16	6	₹14.98
Upper Primary	74.9%	21	6	₹23.01
Secondary	70.9%	25	7	₹41.36
Higher onduary	Sec-57.9%	37	12	₹91.78

Secondary Infrastructure & Fleet Summary

- Teachers Required: 78

Girls Hostels: 0

Cyclone Retrofits: 0
- Transit Minibuses: 6

Transit Feeder Vans: 6

Opex: ₹0.48 Cr/yr

Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Sonepur	69.6%	4	1	₹6.33
Dunguripali	70%	4	1	₹6.33
Birmaharajpur	70.3%	4	1	₹6.33
Tarbha	70.7%	4	1	₹6.33
Binika	71%	5	2	₹9.71

Sundargarh District

Northern Tribal | Vulnerability: 74/100 | GPI: 0.91 | Hill Cost Index: 1.265

SUNDARGARH DISTRICT

SECONDARY EDUCATION SPATIAL CATCHMENT MASTERPLAN

Tribal / Industrial | 17 CD Blocks | Walking Friction: 2.47x (Tobler) | PWD Hill Cost Index: 1.26x

● Existing High Schools (272) ● Proposed Upgrades (140) ★ Greenfield Campuses (94) ▲ Transit Fleet Hubs (109) 5 km Walking Buffer

Baseline Coverage: 45.3% – Target: 97.3% | Allocations: 140 Upgrades, 94 Greenfield, 109 Transit Hubs | Capital Outlay: ₹473.37 Cr

Strategic Directives

Sundargarh (Northern Tribal, Vulnerability: 74/100) comprises 17 CD blocks spanning 9,712 sq.km. With a terrain walking friction factor of 2.47x (Tobler index), a PWD Hill Cost Index of 1.265x, and a Gender Parity Index of 0.91, baseline secondary access is 42.1%. The masterplan allocates 137 high school upgrades, 91 greenfield campuses, 107 student transport hubs (165 mini-buses, 90 vans), 58 dedicated girls' hostels, and 638 subject teacher recruitments. Total estimated capital outlay is Rs. 460.28 Crores, raising universal secondary coverage to 96.5% across all 17 blocks.

Multi-Tier Physical Targets & Capital Outlay

Tier	Base%	Upgr	New	Cost Cr
Primary	64.2%	81	54	₹130.75
Upper Primary	56.2%	103	66	₹220.85
Secondary	42.1%	137	91	₹460.28
Higher ondary	Sec-30.6%	165	110	₹796.01

Secondary Infrastructure & Fleet Summary

Teachers Required: 638

Transit Minibuses: 165

Girls Hostels: 58

Transit Feeder Vans: 90

Cyclone Retrofits: 0

Opex: ₹10.61 Cr/yr

Priority Block Allocations (Top 5 Vulnerable)

Block	Base	Upgr	New	Cost
Subdega	40.4%	9	6	₹30.3
Bishra	40.7%	9	6	₹30.3
Kutra	40.7%	10	6	₹31.37
Gurundia	41%	8	6	₹29.22
Kuarmunda	41%	8	5	₹25.83

Decision Support Framework: Operations Research & Spatial Optimization

Page 33 of 37

5. Operations & Logistics: Phased Rollout, Transit & Staffing

To achieve universal secondary schooling equity efficiently across all 314 blocks, the masterplan establishes a structured 3-phase rollout prioritizing high-vulnerability tribal corridors in Years 1–2:

Phase 1 (Years 1–2): High Vulnerability Tribal Corridors Targeting 9 predominantly southern and western tribal districts. Represents 45% of the capital outlay (₹3119 Cr). Yields the largest marginal gain of +16.5% coverage.

Phase 2 (Years 3–4): Mineral Belts & Plateaus Targeting 11 central and northern districts. Represents 35% of the capital outlay. Adds +11.2% coverage.

Phase 3 (Year 5): Coastal Deltas Targeting 10 developed coastal districts. Represents 20% of the outlay for cyclone retrofits and urban consolidation. Adds +4.8% coverage.

Transit Fleet Architecture

In habitations where physical construction is infeasible, the masterplan deploys a synchronized transit network:

- **1,032 Mini-Buses (24 seats):** Dedicated to all-weather radial trunk routes at ₹4.80 Lakhs/year opex.
- **608 Feeder Vans (12 seats):** Assigned to steep gradient ghat access corridors at ₹3.00 Lakhs/year opex.
- **Annual Fleet Opex:** Calibrated at ₹147.7 Cr/year.

Teacher Retention Cadre & Land Protocols

1. **Tribal Hardship Allowance:** Teachers in high-friction zones receive a 25% Remote Area Allowance with a 3-year bond.
2. **PESA Fast-Track Land Resolution:** Greenfield site selections prioritize unencumbered revenue wasteland with Gram Sabha consent.
3. **Dedicated Residential Hostels:** Construction of 588 units (100 beds each) co-located with high schools.

6. Econometric Returns & Statutory Sign-Off

Econometric Return on Investment (GSDP Contribution)

- **Cohort Impact:** Modeling secondary education completion over a 5-year rollout projects **184,000 students saved from dropout**.
- **Net Present Value (NPV):** Applying a discounted lifetime wage premium of ₹42,000/year (incorporating a 0.70x informal labor market absorption factor and 6% discount rate), the cumulative net present value contribution to Odisha Gross State Domestic Product (GSDP) is estimated at **₹5323 Crores**.
- **Benefit-Cost Ratio:** Against the total secondary capital outlay of ₹6931.8 Crores, the intervention yields a **0.77 BCR**.

Binding State Cabinet Directives

1. **Creation of Specialized Cadre:** Sanction secondary teacher posts with 25% hardship allowances.
2. **Statutory Land Transfers:** Authorize District Collectors to execute zero-cost inter-departmental wasteland transfers under PESA 1996.
3. **Transit Operations by Mission Shakti:** Execute a statewide MoU with Women SHGs for operating feeder transit services.

Operations Research Verification & Official Sign-Off

MATHEMATICAL VERIFICATION CERTIFICATE	STATUTORY AUDIT STATUS
The submodular Pareto MCLP model has been audited against the Nemhauser-Wolsey optimality bounds, guaranteeing $(1 - \frac{1}{e}) \approx 63.2\%$ of global optimal coverage. All 314 CD block catchment buffers and PWD hill cost adjustments have been mathematically certified.	Approved for State Cabinet Consideration Planning Department File: SAMS-SPATIAL-2026-V5 Bhubaneswar, Odisha September 2026

Principal Secretary Department of School & Mass Education	Engineer-in-Chief Public Works Department (PWD)	Development Commissioner Planning & Convergence Department
---	---	--

Appendix: 314 CD Block Regis-
ter

Angul

Block	Vuln	Base%	New	Cost
Angul	70	43.5	12	60.59
Athmallik	69	43.9	12	60.59
Banarpal	71	43.1	13	65.06
Chhendipada	62	46.3	10	50.6
Kaniha	62	46.3	11	56.13
Kishorenagar	73	42.5	11	55.05
Pallahara	63	46	12	59.22
Talcher	69	43.9	10	50.3

Balangir

Block	Vuln	Base%	New	Cost
Agalpur	74	66.1	2	11.46
Balangir	74	66.1	2	10.59
Bangomunda	69	67.8	2	11.46
Belpara	78	64.7	2	10.59
Deogaon	76	65.4	2	9.71
Gudvella	75	65.8	2	10.59
Khaprakhhol	70	67.5	2	10.59
Loisingha	78	64.7	2	10.59
Muribahal	74	66.1	2	10.59
Patnagarh	80	64	2	10.59
Puintala	76	65.4	2	11.46
Saintala	71	67.2	2	9.71
Titilagarh	70	67.5	2	9.71
Turekela	69	67.8	2	10.59

Balasore

Block	Vuln	Base%	New	Cost
Bahanaga	46	75.9	1	4.57
Balasore	41	77.7	1	4.57
Baliapal	38	78.7	1	5.45
Basta	46	75.9	1	5.45
Bhograi	47	75.5	1	5.45
Jaleswar	47	75.5	1	5.45
Khaira	41	77.7	1	4.57
Nilgiri	39	78.3	1	5.45
Oupada	36	79.4	1	4.57
Remuna	39	78.3	1	4.57
Simulia	41	77.7	1	5.45
Soro	42	77.3	1	5.45

Bargarh

Block	Vuln	Base%	New	Cost
Ambabhona	54	73.1	2	9.71
Attabira	58	71.7	2	9.71
Bargarh	57	72	2	10.59
Barpali	60	71	2	9.71
Bhatli	53	73.5	1	6.33
Bheden	58	71.7	2	10.59
Bijepur	54	73.1	2	9.71
Gaisilet	63	70	2	10.59
Jharbandh	53	73.5	1	6.33
Padampur	54	73.1	1	6.33
Paikmal	55	72.8	2	10.59
Sohela	60	71	1	7.2

Bhadrak

Block	Vuln	Base%	New	Cost
Basudevpur	44	76.6	1	5.45
Bhadrak	45	76.2	1	5.45
Bhandaripokhari	39	78.3	1	5.45
Bonth	40	78	1	5.45
Chandabali	40	78	1	4.57
Dhamnagar	40	78	1	5.45
Tihidi	43	77	1	5.45

Boudh

Block	Vuln	Base%	New	Cost
Boudh	76	41.4	14	70.89

Block	Vuln	Base%	New	Cost
Harabhanga	69	43.9	16	81.97
Kantamal	72	42.8	15	75.06

Cuttack

Block	Vuln	Base%	New	Cost
Athagarh	31	81.2	1	3.7
Badamba	34	80.1	1	4.57
Banki	33	80.5	1	4.57
Banki-Dampada	37	79	1	4.57
Baranga	36	79.4	1	4.57
Cuttack Sadar	38	78.7	1	4.57
Kantapada	31	81.2	1	4.57
Mahanga	30	81.5	1	3.7
Narasinghpur	34	80.1	1	4.57
Niali	40	78	1	4.57
Nischintakoili	39	78.3	1	4.57
Salepur	35	79.8	1	4.57
Tangi-Choudwar	39	78.3	1	4.57
Tigiria	32	80.8	1	4.57

Deogarh

Block	Vuln	Base%	New	Cost
Barkote	68	44.2	12	60.59
Reamal	63	46	12	61.97
Tileibani	69	43.9	14	70.89

Dhenkanal

Block	Vuln	Base%	New	Cost
Bhuban	58	47.7	7	35.84
Dhenkanal	52	49.8	6	30.3
Gondia	52	49.8	6	31.37
Hindol	50	50.5	6	30.3
Kamakhyanagar	57	48	6	30.3
Kankadahad	55	48.8	7	34.76
Odapada	54	49.1	8	40.29
Parjang	57	48	8	40.29

Gajapati

Block	Vuln	Base%	New	Cost
Gumma	85	38.2	10	49.22
Kashinagar	88	37.2	10	50.3
Mohana	88	37.2	8	40.59
Nuagada	86	37.9	9	46.13
Parlakhemundi	91	36.2	8	40.29
R.Udayagiri	93	35.5	8	41.67
Rayagada	83	39	9	46.13

Ganjam

Block	Vuln	Base%	New	Cost
Aska	63	70	1	7.2
Beguniapada	61	70.7	1	6.33
Bellaguntha	57	72	1	6.33
Bhanjanagar	62	70.3	1	7.2
Buguda	53	73.5	1	6.33
Chatrapur	64	69.6	1	7.2
Chikiti	54	73.1	1	6.33
Dharakote	58	71.7	1	6.33
Digapahandi	53	73.5	1	6.33
Ganjam	57	72	1	5.45
Hinjilicut	59	71.3	1	6.33
Jagannathprasad	53	73.5	1	6.33
Kabisuryanagar	55	72.8	1	6.33
Khallikote	58	71.7	1	5.45
Kukudakhandi	53	73.5	1	6.33
Patrapur	61	70.7	1	5.45
Polasara	57	72	1	6.33
Purushottampur	58	71.7	1	5.45
Rangeilunda	55	72.8	1	6.33
Sanakhemundi	55	72.8	1	6.33
Sheragada	60	71	1	6.33
Surada	64	69.6	1	6.33

Jagatsinghpur

Block	Vuln	Base%	New	Cost
Balikuda	42	77.3	1	3.7
Biridi	38	78.7	1	3.7
Erasama	37	79	1	3.7
Jagatsinghpur	37	79	1	3.7
Kujang	37	79	1	3.7
Naugaon	34	80.1	1	3.7
Raghunathpur	39	78.3	1	3.7
Tirtol	41	77.7	1	3.7

Jajpur

Block	Vuln	Base%	New	Cost
Barchana	47	75.5	1	4.57
Bari	45	76.2	1	4.57
Binjharpur	42	77.3	1	4.57
Danagadi	50	74.5	1	5.45
Dasarathpur	47	75.5	1	5.45
Dharmasala	39	78.3	1	4.57
Jajpur	42	77.3	1	5.45
Korei	42	77.3	1	4.57
Rasulpur	43	77	1	4.57
Sukinda	44	76.6	1	5.45

Jharsuguda

Block	Vuln	Base%	New	Cost
Jharsuguda	50	74.5	1	6.33
Kirmira	49	74.8	1	5.45
Kolabira	47	75.5	1	6.33
Laikera	55	72.8	1	6.33
Lakhanpur	51	74.2	1	6.33

Kalahandi

Block	Vuln	Base%	New	Cost
Bharmagarh	84	38.6	7	34.76
Bhawanipatna	80	40	7	34.76
Golamunda	88	37.2	8	40.29
Jaipatna	87	37.5	8	40.29
Junagarh	84	38.6	7	35.84
Kalampur	89	36.9	7	35.84
Karlamunda	82	39.3	9	44.76
Kesinga	88	37.2	9	44.76
Koksara	87	37.5	8	39.22
Lanjigarh	84	38.6	8	41.67
Madanpur Rampur	81	39.6	8	40.29
Narla	83	39	7	34.76
Th.Rampur	82	39.3	8	40.29

Kandhamal

Block	Vuln	Base%	New	Cost
Baliguda	89	36.9	8	40.29
Chakapada	94	35.1	9	44.76
Daringbadi	97	34	10	50.3
G.Udayagiri	93	35.5	10	50.6
K.Nuagaon	92	35.8	9	46.13
Khajuripada	95	34.8	8	40.29
Kotagarh	89	36.9	9	46.13
Phiringia	89	36.9	7	36.14
Phulbani	94	35.1	10	50.6
Raikia	91	36.2	7	36.14
Tikabali	91	36.2	8	40.59
Tumudibandha	87	37.5	10	50.3

Kendrapara

Block	Vuln	Base%	New	Cost
Aul	36	79.4	1	4.57
Derabish	36	79.4	1	4.57
Garadapur	34	80.1	1	4.57
Kendrapara	36	79.4	1	4.57
Mahakalapada	40	78	1	4.57
Marshaghai	43	77	1	4.57
Pattamundai	39	78.3	1	4.57
Rajakanika	42	77.3	1	4.57
Rajnagar	35	79.8	1	4.57

Kendujhar

Block	Vuln	Base%	New	Cost
Anandapur	83	39	8	41.67
Bansapal	72	42.8	8	40.29
Champua	83	39	8	41.67
Ghasipura	81	39.6	7	34.76
Ghatgaon	73	42.5	8	40.29
Harichandanpur	74	42.1	8	40.29
Hatadihi	75	41.8	8	40.29
Jhumpi	75	41.8	6	31.37
Joda	72	42.8	6	31.37
Keonjhar	82	39.3	9	44.76
Patna	79	40.4	7	35.84
Saharpada	78	40.7	8	41.67
Telkoi	83	39	7	34.76

Khordha

Block	Vuln	Base%	New	Cost
Balianta	30	81.5	1	3.7
Balipatna	33	80.5	1	4.57
Banapur	31	81.2	1	3.7
Begunia	30	81.5	1	3.7
Bhubaneswar	30	81.5	1	4.57
Bolagarh	32	80.8	1	4.57
Chilika	30	81.5	1	3.7
Jatni	30	81.5	1	4.57
Khordha	30	81.5	1	4.57
Tangi	30	81.5	1	3.7

Koraput

Block	Vuln	Base%	New	Cost
B. Singhpur	93	35.5	7	35.84
Bandhugaon	96	34.4	7	35.84
Boipariguda	86	37.9	8	40.29
Borigumma	94	35.1	8	40.29
Dasamantapur	96	34.4	7	35.84
Jeypore	94	35.1	8	40.29
Koraput	91	36.2	8	40.29
Kotpad	87	37.5	7	34.76
Kundra	87	37.5	6	31.37
Lamtaput	92	35.8	7	35.84
Laxmipur	87	37.5	7	34.76
Machhkund	92	35.8	8	40.29
Nandapur	93	35.5	7	34.76
Narayanpatna	96	34.4	8	40.29

Malkangiri

Block	Vuln	Base%	New	Cost
Chitrakonda	96	34.4	12	59.22
Kalimela	91	36.2	14	70.59
Khairput	99	33.4	12	60.59
Korkunda	89	36.9	12	60.59
Malkangiri	89	36.9	12	60.59
Mathili	95	34.8	12	60.59
Podia	98	33.7	12	59.52

Mayurbhanj

Block	Vuln	Base%	New	Cost
Badasahi	87	37.5	3	15.84
Bahalda	78	40.7	3	15.84
Bangiriposi	79	40.4	3	15.84
Baripada	76	41.4	3	15.84
Barsahi	87	37.5	3	15.84
Betnoti	79	40.4	4	20.3
Bijatala	86	37.9	3	15.84
Bisoi	86	37.9	4	20.3
Gopabandhunagar	87	37.5	4	20.3
Jamda	85	38.2	3	15.84
Jashipur	87	37.5	3	15.84
Kaptipada	77	41	3	15.84
Karanjia	83	39	4	20
Khunta	83	39	3	15.84
Kusumi	86	37.9	4	20.3
Morada	81	39.6	3	15.84
Rairangpur	85	38.2	4	20
Rasgovindpur	84	38.6	4	20
Samakhunta	76	41.4	3	14.46
Saraskana	86	37.9	4	20
Sukruli	82	39.3	3	15.84
Suliapada	79	40.4	4	20.3

Block	Vuln	Base%	New	Cost
Thakurmunda	86	37.9	4	20.3
Tiring	78	40.7	3	15.84
Tirto	86	37.9	3	15.84
Udala	77	41	4	20.3

Nabarangpur

Block	Vuln	Base%	New	Cost
Chandahandi	91	36.2	9	46.13
Dabugam	85	38.2	7	34.76
Jharigam	86	37.9	7	35.84
Kosagumuda	93	35.5	8	40.29
Nabarangpur	87	37.5	9	46.13
Nandahandi	86	37.9	7	36.14
Papuhandi	86	37.9	8	40.59
Raighar	87	37.5	7	36.14
Tentulikhunti	89	36.9	7	35.84
Umerkote	84	38.6	8	40.59

Nayagarh

Block	Vuln	Base%	New	Cost
Bhapur	44	76.6	1	6.33
Dasapalla	46	75.9	1	7.2
Gania	45	76.2	1	6.33
Khandapada	48	75.2	2	9.71
Nayagarh	49	74.8	2	9.71
Nuagaon	51	74.2	2	9.71
Odagaon	42	77.3	1	6.33
Ranpur	50	74.5	2	9.71

Nuapada

Block	Vuln	Base%	New	Cost
Boden	85	62.2	4	19.42
Khariar	91	60.2	4	21.18
Komna	90	60.5	4	19.42
Nuapada	88	61.2	4	21.18
Sinapali	87	61.5	4	19.42

Puri

Block	Vuln	Base%	New	Cost
Astaranga	37	79	1	4.57
Brahmagiri	33	80.5	1	4.57
Delanga	34	80.1	1	4.57
Gop	38	78.7	1	4.57
Kakatpur	40	78	1	5.45
Kanas	37	79	1	4.57
Krushnaprasad	36	79.4	1	4.57
Nimapada	33	80.5	1	4.57
Pipili	31	81.2	1	4.57
Puri Sadar	36	79.4	1	5.45
Satyabadi	33	80.5	1	4.57

Rayagada

Block	Vuln	Base%	New	Cost
Bissamacuttack	89	36.9	9	44.76
Chandrapur	92	35.8	10	50.3
Gudari	93	35.5	8	40.59
Gunupur	92	35.8	10	51.67
Kalyansingpur	91	36.2	10	51.67
Kashipur	89	36.9	8	40.29
Kolnara	91	36.2	11	54.75
Muniguda	89	36.9	9	46.13
Padmapur	92	35.8	9	46.13
Putasing	85	38.2	9	46.13
Rayagada	88	37.2	10	50.6

Sambalpur

Block	Vuln	Base%	New	Cost
Bamra	67	68.5	3	16.91
Dhankauda	64	69.6	3	15.74
Jamankira	67	68.5	3	15.74
Jujomura	62	70.3	3	15.74
Kuchinda	64	69.6	3	14.86
Maneswar	65	69.2	3	14.86
Naktideul	58	71.7	3	14.86

Block	Vuln	Base%	New	Cost
Rairakhol	60	71	2	11.46
Rengali	58	71.7	3	13.98

Subarnapur

Block	Vuln	Base%	New	Cost
Binika	60	71	2	9.71
Birmaharajpur	62	70.3	1	6.33
Dunguripali	63	70	1	6.33
Sonepur	64	69.6	1	6.33
Tarbha	61	70.7	1	6.33
Ullunda	58	71.7	1	6.33

Sundargarh

Block	Vuln	Base%	New	Cost
Balisankara	71	43.1	5	25.83
Bargaon	73	42.5	6	30.3
Bishra	78	40.7	6	30.3
Bonei	75	41.8	6	29.22
Gurundia	77	41	6	29.22
Hemgir	74	42.1	4	21.38
Koida	68	44.2	6	30.3
Kuarmunda	77	41	5	25.83
Kutra	78	40.7	6	31.37
Lahunipara	73	42.5	5	24.46
Lathikata	77	41	5	24.76
Lephripara	74	42.1	5	25.83
Nuagaon	73	42.5	5	24.76
Rajgangpur	76	41.4	5	25.83
Subdega	79	40.4	6	30.3
Sundargarh	74	42.1	5	24.76
Tangarpali	72	42.8	5	25.83